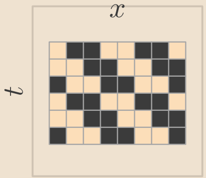
Selecting a relative-periodic domain with Bernoulli NML

1 Observed spacetime

Choose a candidate
period and shift.
Here $p = 2$ and $s = 1$.



The observed CA spacetime U
is the pattern to be explained.

2 Orbit classes

Translate by $(t, x) \mapsto (t + p, x + s)$.
Each orbit gives
one template bit.

1	2	3	4	1	2
4	1	2	3	4	1
3	4	1	2	3	4
2	3	4	1	2	3
1	2	3	4	1	2

The orbit partition shrinks
the spacetime block
to a smaller template.

3 Majority-vote fit

Count ones and zeros
on each orbit.
Majority vote sets
the template bit.

orbit	count	vote
1	5 / 1	→ ■
2	1 / 5	→ □
3	4 / 2	→ ■
4	2 / 4	→ □

The voted bits form $B^*(p, s)$,
and $M = U \oplus B^*(p, s)$
is the residual mask.

4 Score and select

Score each candidate
with Bernoulli NML.
Smaller totals are better.

NML(p, s) = fit + penalty

Fit pure orbits reduce fit

Penalty extra bits raise cost

Choose the candidate with the smallest total NML.

Output: the selected scaffold and its residual mask.